

3.2.5 Using and working with materials

In relation to **at least one** material category or system, students should know and understand in addition to material properties, the factors listed below.

Properties of materials

Content	Potential links to maths and science
Students must know and understand how different properties of materials and components are used in commercial products, how properties influence use and how properties affect performance. Students must know and understand the physical and mechanical properties relevant to commercial products in their chosen area as follows. • Papers and boards (flyers/leaflets and card based food packaging). • Timber based materials (traditional timber children's toys and flat pack furniture). • Metal based materials (cooking utensils and hand tools). • Polymers (polymer seating and electrical fittings). • Textile based materials (sportswear and furnishings). • Electronic and mechanical systems (motor vehicles and domestic appliances).	How physical and working properties are selected related and used in commercial products when designing and making.

The modification of properties for specific purposes

Content	Potential links to maths and science
• Additives to prevent moisture transfer (paper and boards). • Seasoning to reduce moisture content of timbers (timber based materials). • Annealing to soften material to improve malleability (metal based materials). • Stabilisers to resist UV degradation (polymers). • Flame retardants reduce combustion and fire hazards (textile based materials). • Photosensitive PCB board in PCB manufacture and anodizing aluminium to improve surface hardness (electronic and mechanical systems).	

How to shape and form using cutting, abrasion and addition

Content	Potential links to maths and science
• Papers and boards (how to cut, crease, score, fold and perforate card). • Timber based materials (how to cut, drill, chisel, sand and plane). • Metal based materials (how to cut, drill, turn, mill, cast, braze and weld). • Polymers (how to cut, drill, cast, deform, print and weld). • Textile based materials (how to sew, pleat, gather, quilt and pipe). • Electronic and mechanical systems (how to cut, drill and solder).	